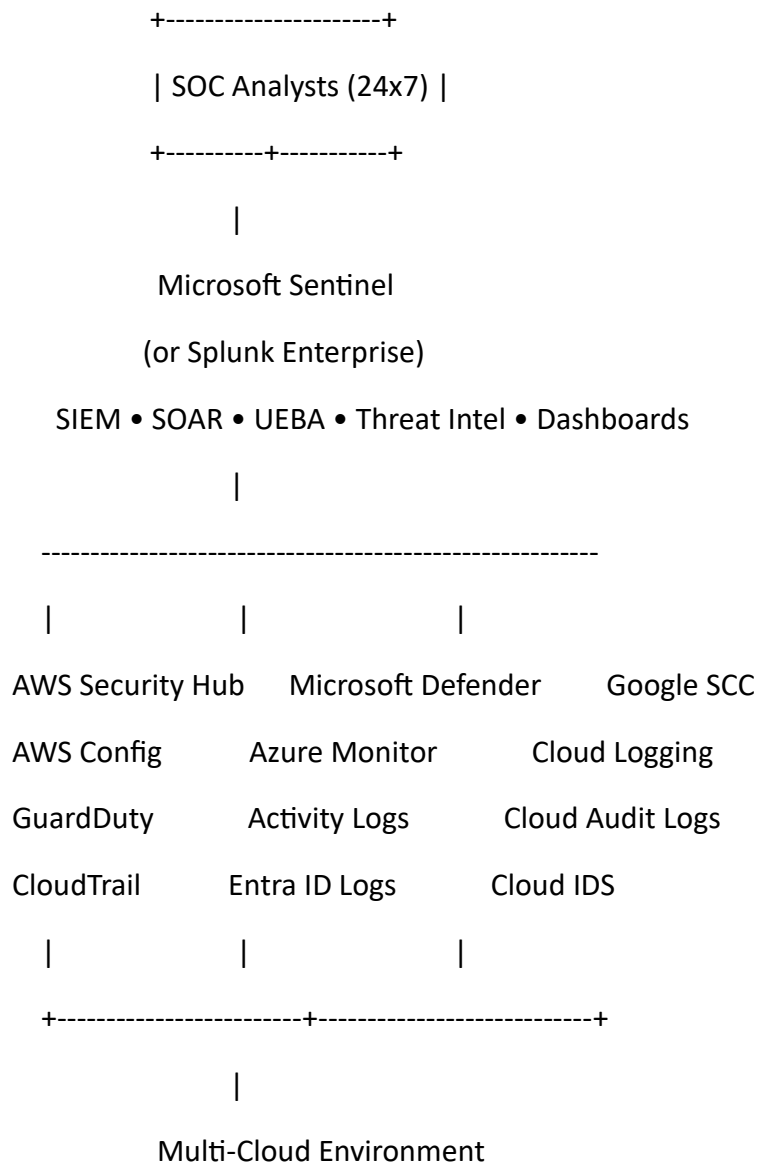


Module 6 – Detection, Monitoring & Incident Response

Objective

Implement a centralized Security Operations capability to provide continuous monitoring, real-time threat detection, and incident response across AWS, Azure, and GCP.

1. Target-State Architecture



2. CSPM Strategy

Cloud CSPM Solution

AWS Security Hub + AWS Config

Azure Microsoft Defender for Cloud

GCP Security Command Center

Continuously monitor:

- Public storage
- Public databases
- IAM changes
- Security groups / NSGs / Firewall rules
- Encryption compliance
- Logging configuration
- Backup settings

Critical findings are automatically forwarded to the SIEM.

3. Telemetry Collection

AWS	Azure	GCP
CloudTrail	Activity Logs	Cloud Audit Logs
GuardDuty	Defender Alerts	SCC Findings
VPC Flow Logs	NSG Flow Logs	VPC Flow Logs
Security Hub	Azure Monitor	Cloud Logging
IAM Logs	Entra ID Logs	IAM Audit Logs

All logs are centralized in **Microsoft Sentinel (or Splunk)** for correlation, investigation, and incident management.

4. Alert Fatigue Management

- Event correlation
- Alert deduplication
- Risk-based prioritization
- UEBA baselines
- Threat intelligence enrichment
- SOAR automation for repetitive tasks

KPIs

- MTTD < 15 minutes
 - MTTR < 60 minutes
 - False Positive Rate < 5%
-

5. Incident Response Scenario

Scenario

A GitLab CI service account with wildcard (*:*) IAM permissions is compromised and used to access a production database.

Detection

- GuardDuty detects suspicious API activity.
- CloudTrail records privileged access.
- Database audit logs show unusual queries.
- SIEM correlates alerts and creates a Critical incident.

Containment

- Disable compromised service account.
- Revoke access tokens.
- Rotate credentials.
- Isolate affected resources.

Eradication

- Remove malicious IAM permissions.

- Review Terraform state.
- Scan for persistence mechanisms.
- Validate environment integrity.

Recovery

- Deploy a new least-privilege OIDC service account.
 - Restore data if required.
 - Re-enable services after validation.
 - Increase monitoring for 72 hours.
-

6. Incident Communication

During the Incident (ITSO)

- Incident confirmed
- Business impact
- Actions taken
- Current risk
- Next update time

Post-Incident Report

- Executive summary
 - Root cause
 - Timeline
 - Impact assessment
 - Lessons learned
 - Corrective actions
-

7. Success Metrics

KPI	Target
MTTD	< 15 min
MTTR	< 60 min
Critical Findings SLA	< 24 hrs
CSPM Compliance	> 95%
Log Coverage	100%
Security Monitoring Availability	99.9%